

COMPASS

Prostate Cancer Consultation Dashboard

Complete System Flow Documentation

Login · Data Pipeline · Surveys · REDCap · DB Schema

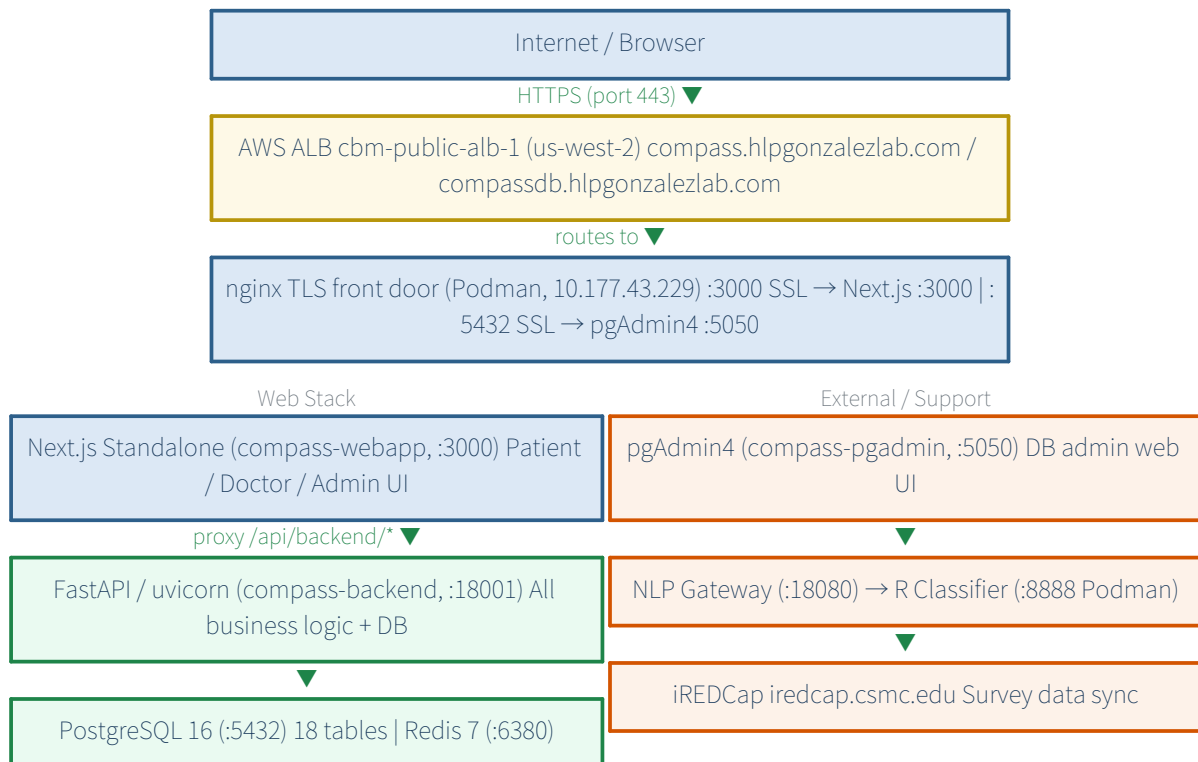
Server	compass (10.177.43.229) — EC2 us-west-2
Web App	https://compass.hlpgonzalezlab.com
DB Admin	https://compassdb.hlpgonzalezlab.com
Branch	production/official
Created	2026-09-18

Color Legend

Color	Meaning
☒ Blue	Web / UI (Browser, Next.js, nginx)
☒ Green	Backend / Database (FastAPI, PostgreSQL)
☒ Orange	External services (REDCap, NLP, Azure OpenAI)
☒ Red	Auth / Security (JWT, cookies, rate limits)
☒ Grey	Behavior tracking / Audit logs

2. System Architecture Overview

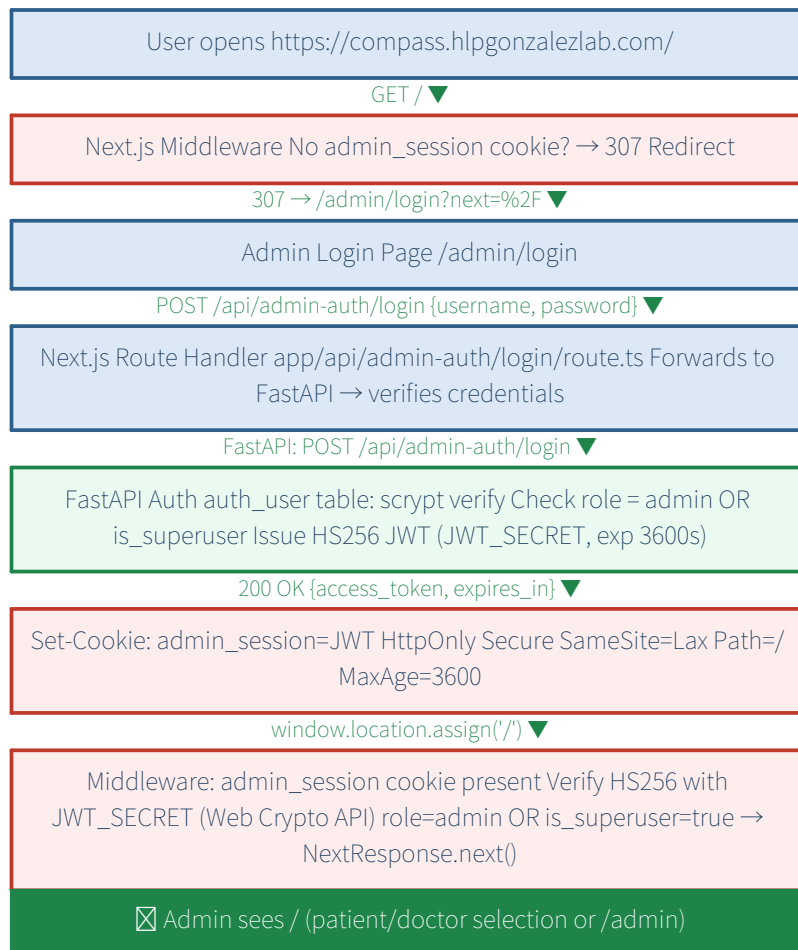
Complete service map — from public internet to internal services.



Systemd Services

Service	Port	Description
compass-webapp	:3000 (127.0.0.1)	Next.js standalone — patient/doctor/admin UI
compass-backend	:18001 (0.0.0.0)	FastAPI/uvicorn — all API endpoints
compass-nginx-tls	:3000/:5432 (SSL)	nginx reverse proxy (Podman, TLS termination)
compass-pgadmin	:5050 (127.0.0.1)	pgAdmin4 — DB management web UI
compass-nlp-gateway	:18080 (0.0.0.0)	NLP gateway — routes to R classifier
compass-nlp-classifier	:8888 (127.0.0.1)	R Random Forest classifiers in Podman
compass-pipeline-watch	(filesystem)	Watches pipeline_drop_dir, triggers NLP+AI

3. Admin Login Flow



DB Tables Touched

Table	Operation	Purpose
auth_user	SELECT	Lookup user by username, verify scrypt hash, check role

Note: logout clears the cookie (`maxAge=0`). No server-side session invalidation.

4. Patient / Doctor Personal Link Access

Patients and doctors access the app via unique personal links — no password required. The link contains AES-SIV hashed tokens (never raw PHI).

Token structure: <hashedPatient>_<hashedDoctor>_<hashedDate>.csv

AES-SIV decryption (DEID_KEY env) → real SID / doctor number / visit week

URL Pattern	View Rendered
/?fileid=X&patid=Y&survey=first-visit	PatientInitialVisitReportV42 (First Visit)
/?fileid=X&patid=Y&survey=follow-up	PatientFollowUpReportV38 (Follow-up Surveys)
/?fileid=X&doctorid=Y	PhysicianReportsModifiedV41Timothy (Doctor)
/?f=<stem>&view=first-report	PatientInitialVisitReportV42 (shortlink)
/ (no params, admin cookie present)	Patient/Doctor Selection Screen
/ (no params, no admin cookie)	→ 307 /admin/login (middleware redirect)

Personal Link URL /?fileid=X&patid;=Y or /?fileid=X&doctorid;=Y

Middleware: hasPersonalLink? → NextResponse.next() (no auth required) ▼

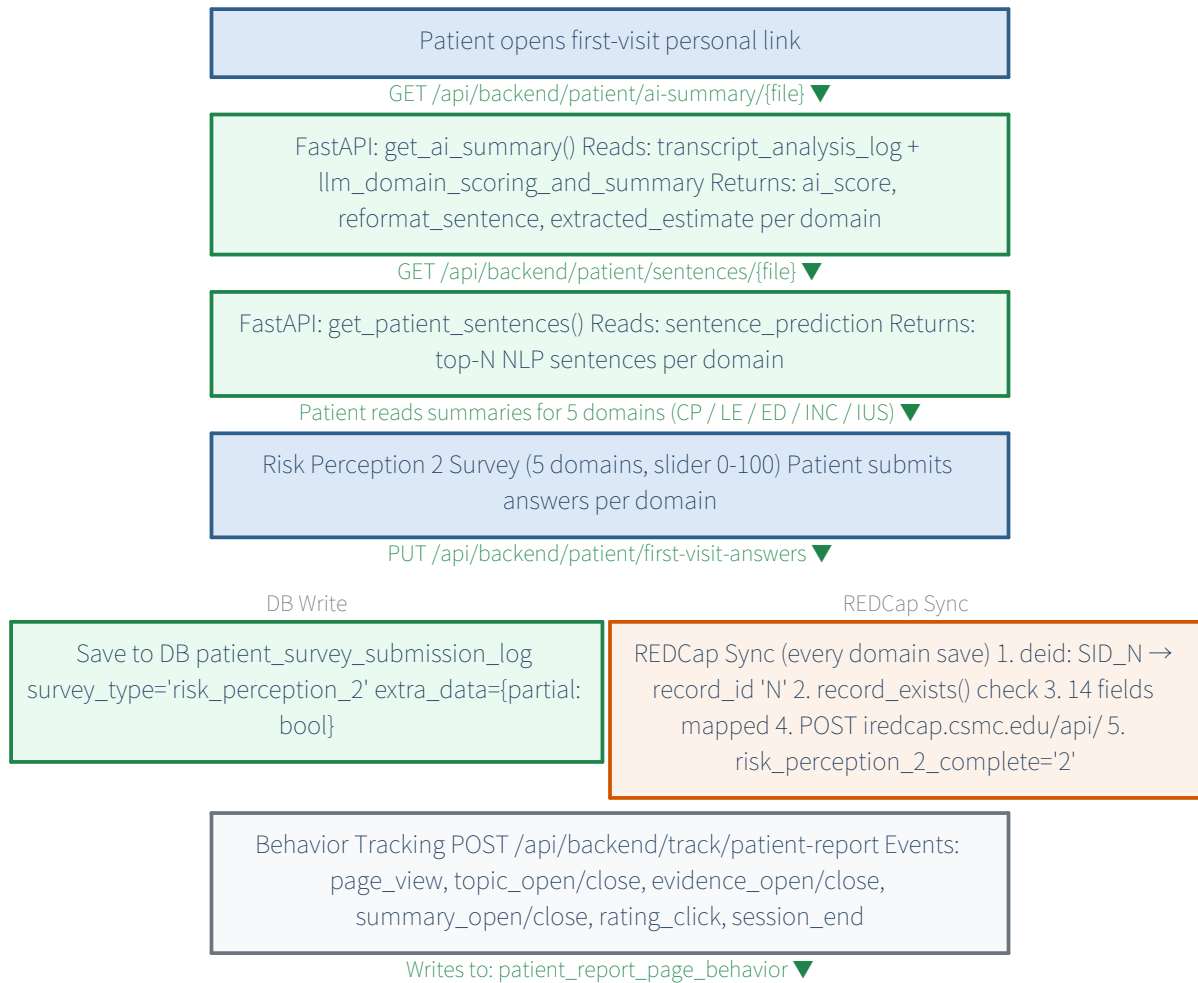
page.tsx Reads searchParams: fileid, patid, doctorid, visit, survey
Determines view type: patient / doctor / selection

X-API-Key injected server-side by /api/backend proxy ▼

FastAPI Backend (:18001) All patient/doctor endpoints require API key
PHI audit: every response logged → phi_access_log

5. Patient First Visit Flow

Patient opens their personal link for the first visit. The app loads AI-generated summaries and presents a Risk Perception survey.

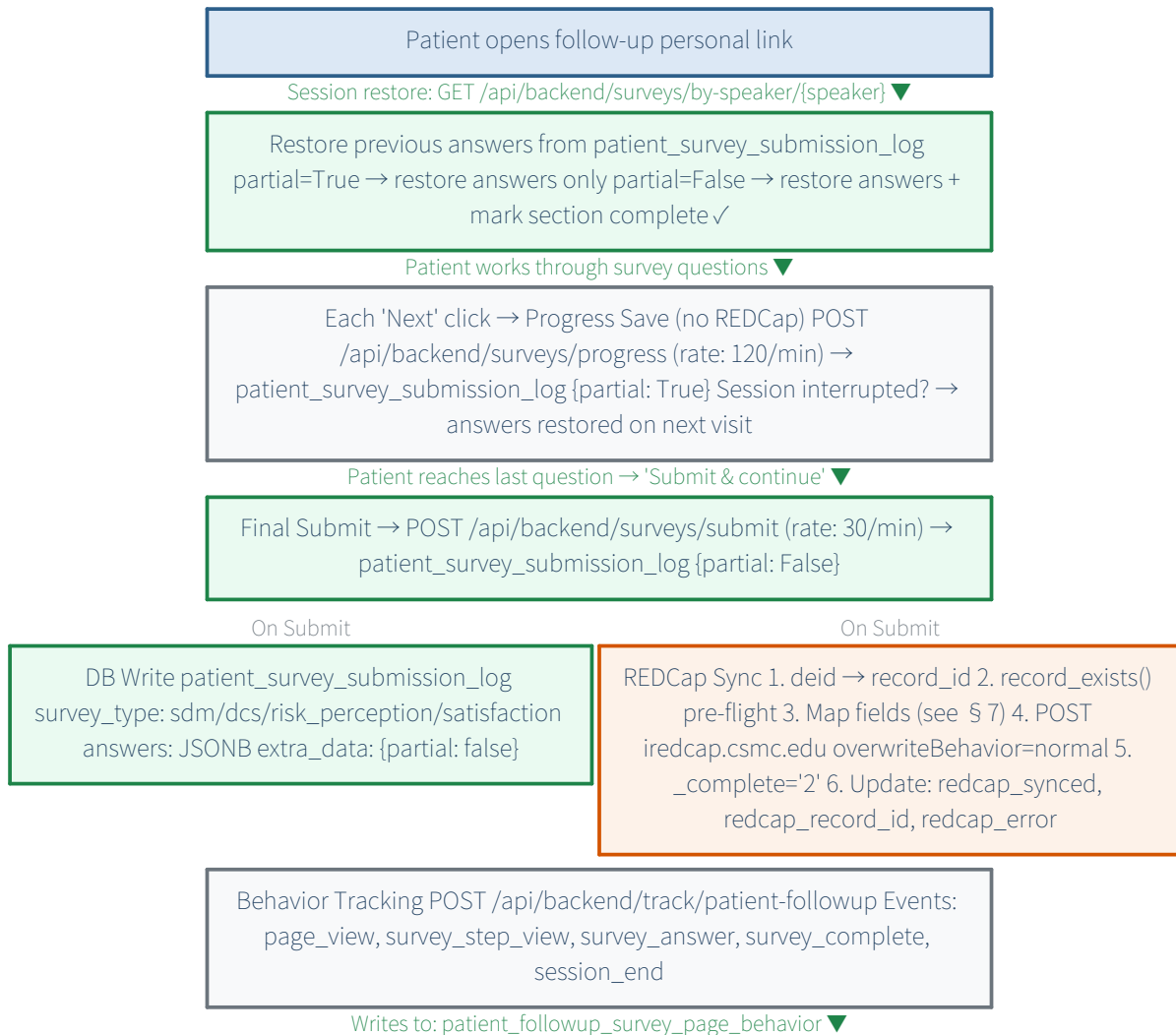


DB Tables

Table	Operation	Purpose
transcript_analysis_log	SELECT	Find analysis run for this file
llm_domain_scoring_and_summary	SELECT	AI scores + patient-facing summaries
sentence_prediction	SELECT	Top-N NLP sentences per domain
patient_summary	SELECT	Parent record existence check
patient_survey_submission_log	INSERT	Store Risk Perception 2 answers
patient_report_page_behavior	INSERT	UI interaction events
phi_access_log	INSERT	HIPAA audit log for each API call

6. Patient Follow-up Survey Flow

4 sequential surveys: SDM (4Q) → DCS (16Q) → Risk Perception (5Q) → Satisfaction (1Q). Forward-only navigation. Progress saved to DB on every Next click (no REDCap). Final Submit triggers REDCap sync exactly once per survey.



7. REDCap Field Mapping

All survey submissions that reach POST /api/surveys/submit trigger REDCap sync if REDCAP_ENABLED=True. Timeout: 90 seconds per call (iredcap.csmc.edu).

Survey	REDCap Instrument	# Fields	Value Transform	Trigger
dcs	decisional_conflict_survey	16	0-based → 1-based (+1)	Final Submit
sdm	shared_decision_making_sdm	4	yes/no → 1/0; scale text → 1-4	Final Submit
risk_perception	risk_perception	5	Slider 0-100 → int	Final Submit
risk_perception_2	post_risk_perception_2	14	checkbox: field___code=1/0	Final Submit
satisfaction	patient_satisfaction	1	Free text (overwrite)	Final Submit
risk_perception_2	risk_perception_2 (1st visit)	14	checkbox: field___code=1/0	Every domain save

DCS Mapping (16 questions):

DCS Frontend Key	REDCap Field	Transform
q1	dcs1_v2	value + 1 (0→1 ... 4→5)
q2	dcs2_v2	value + 1 (0→1 ... 4→5)
q3	dcs3_v2	value + 1 (0→1 ... 4→5)
q4	dcs4_v2	value + 1 (0→1 ... 4→5)
q5	dcs5_v2	value + 1 (0→1 ... 4→5)
q6	dcs6_v2	value + 1 (0→1 ... 4→5)
q7	dcs7_v2	value + 1 (0→1 ... 4→5)
q8	dcs8_v2	value + 1 (0→1 ... 4→5)
q9 ... q16	dcs9_v2 ... dcs16_v2	value + 1

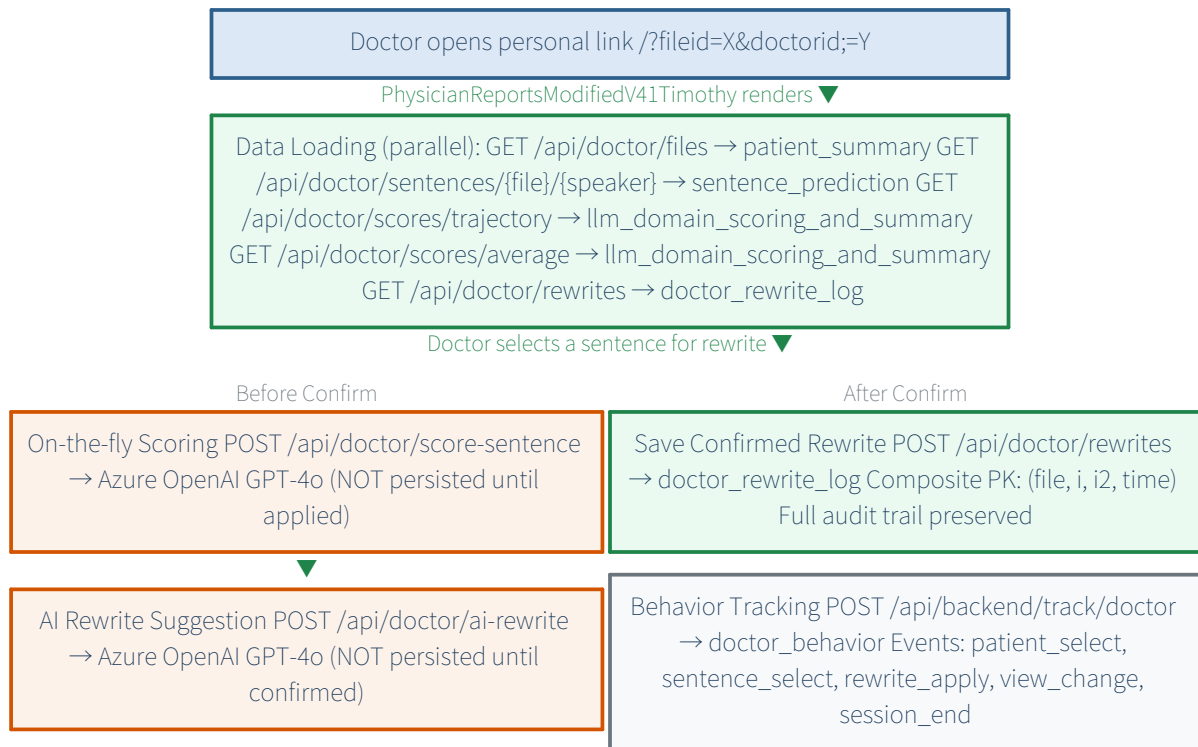
SDM Mapping:

SDM Frontend Key	REDCap Field	Value
q1 (options aware)	sdmp_options	yes→1, no→0
q2 (pros/cons)	sdm_ptos	yes→1, no→0
q3 (discussed cons)	sdm_cons	yes→1, no→0
q4 (preference)	sdm_pref	scale text → 1-4

Risk Perception Mapping:

Risk Perception Frontend Key	REDCap Field
cancerRiskUntreated	risk_percep_1_1
cancerRiskTreated	risk_percept2_2
erectileDysfunctionRisk	risk_percept_3_3
urinaryIncontinenceRisk	risk_percept_4_4
irritativeUrinaryRisk	risk_percep_5_5

8. Doctor Dashboard Flow

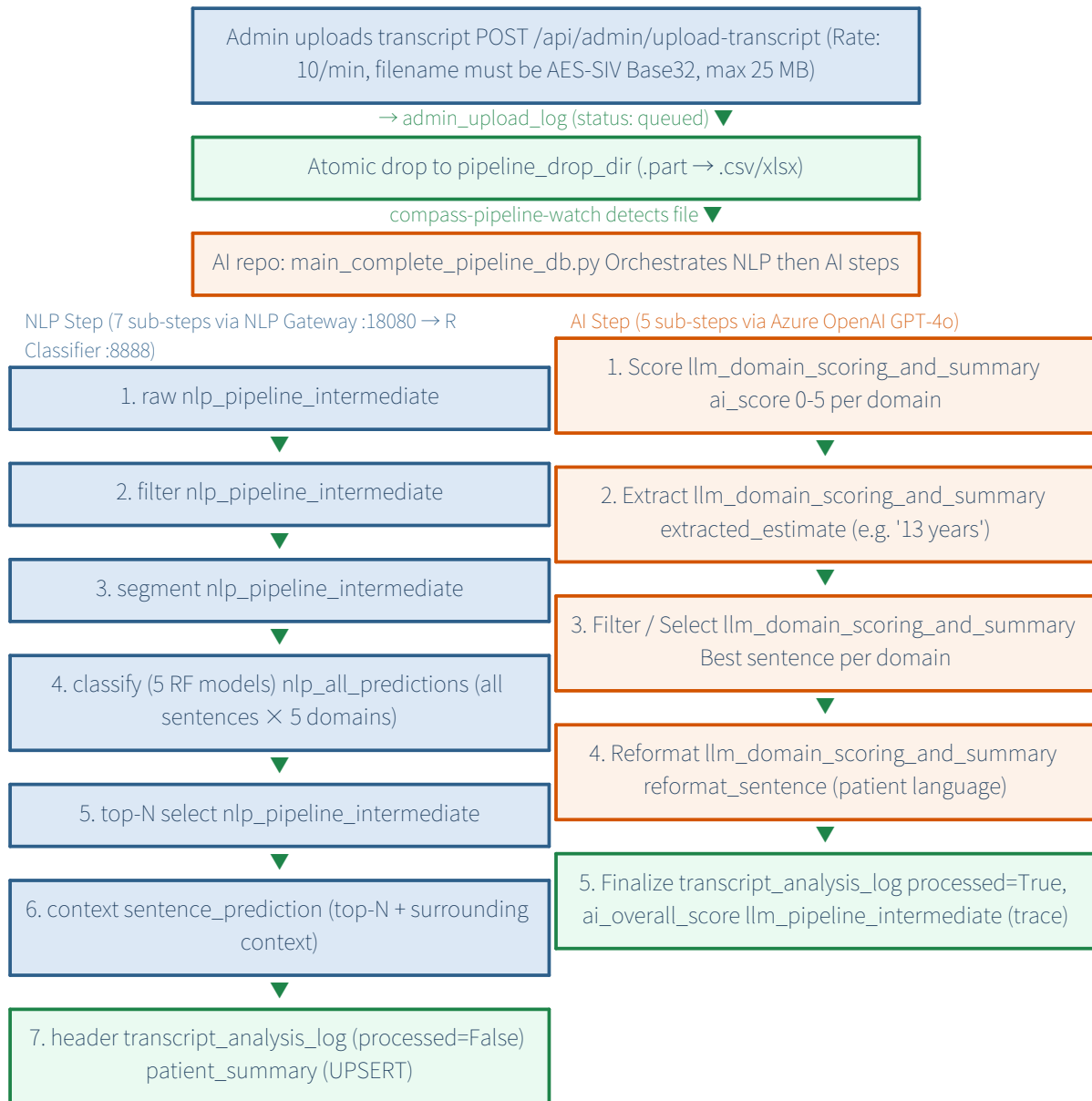


DB Tables

Table	Operation	Purpose
patient_summary	SELECT	List available patient files
sentence_prediction	SELECT	Top-N NLP sentences per domain
llm_domain_scoring_and_summary	SELECT	ai_score, visit week for trajectory chart
doctor_rewrite_log	SELECT / INSERT	Rewrite history (audit trail, composite PK)
doctor_behavior	INSERT	UI interaction events
phi_access_log	INSERT	HIPAA audit for each API call

9. Data Pipeline Flow (NLP + AI)

The pipeline runs outside the webapp. Admin uploads a de-identified transcript CSV/XLSX. compass-pipeline-watch detects it and calls the AI repo orchestrator.



10. Database Schema — 18 Tables

Auth (2 tables)

Table	PK	Key Columns	Purpose
auth_user	id	username, password_hash (bcrypt), role, is_superuser	Admin user accounts
auth_api_key	id	key_hash, user_id (FK), expires_at, last_used_at	API keys for patient/doctor endpoints

Pipeline (7 tables)

Table	Purpose
transcript_analysis_log	One row per pipeline run — metadata, xlsx binary, timing, scores (parent)
patient_summary	One row per patient file+speaker — FK anchor for surveys
sentence_prediction	Top-N NLP sentences per domain per run
nlp_all_predictions	ALL sentences × 5 domain probabilities (full NLP output)
nlp_pipeline_intermediate	JSONB snapshots of NLP steps (raw/filtered/sentences/top_by_model)
llm_domain_scoring_and_summary	GPT-4o output: ai_score, extracted_estimate, reformat_sentence per domain
llm_pipeline_intermediate	Per-domain LLM input/output trace (written by AI repo)

Survey Submission (1 table)

Table	Key Columns	Purpose
patient_survey_submission_log	file, speaker, survey_type, answers (JSONB), extra_data {partial: bool}, sid, doctor, redcap_synced, redcap_record_id, redcap_error	ALL survey answers (first-visit + follow-up). Append-only. Tracks REDCap sync state.

Behavior Tracking (3 tables)

Table	Key Events	Purpose
patient_report_page_behavior	page_view, topic_open/close, evidence_open, saving_click, session_end	First-visit clickstream
patient_followup_survey_page_behavior	survey_step_view, survey_answer, survey_complete, session_end	Follow-up survey engagement
doctor_behavior	patient_select, sentence_select, rewrite_apply, document_click, document_undo	Doctor dashboard interactions

Admin / Audit (3 tables)

Table	Purpose
admin_upload_log	Records each admin transcript upload attempt (queued_filename, status, uploaded_by)
session_recording	rrweb session replay chunks — gzipped JSON, tagged by area (patient_first/followup/doctor)
phi_access_log	HIPAA 164.312(b) audit trail — every request returning patient data (actor, IP, path, status)